Finding More Nucleus Centers

Nuclear instance segmentation: stage report 7

bdd26 research record

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Abstract

This report reviews point-localization studies P17 through P26. P25 adapted all encoder layers with a layerwise learning rate. Both seeds improved overall recall and precision and detected a net seven and ten more Dead nuclei, respectively. Neither met the original minimum requirements for Dead recall and precision, and neither paired interval had a strictly positive lower bound. We will not pursue this particular training configuration further. P26 anchors trainable weights to their pretrained values; both seeds fail the original gate.

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1 The task and how to read the results

The project ultimately aims to identify each nucleus in microscopy images, delineate its boundary, and classify it. This stage isolates an earlier step: predict a center for each nucleus. A correct center is useful for constructing an instance, but does not establish a correct boundary or class.

Dead denotes dying or dead nuclei, including apoptotic and necrotic examples. These are rare and often difficult to locate. Predictions are matched one-to-one to annotated centers using a fixed rule. **Recall** is the fraction of annotated nuclei found; **precision** is the fraction of predicted points that match an annotation. Producing more points can recover misses but can also create false alarms, so both numbers matter.

The main point head predicts locations, not a class for each point. Dead recall is calculated after matching, using the ground-truth Dead labels to select the subgroup; precision is the matched fraction of all main-map points, not Dead classification precision. An interior nucleus has a mask that does not touch the image boundary. A small Dead nucleus is an interior Dead object with an annotated mask area below 67 native pixels. One-to-one matching allows a Euclidean distance of at most three native pixels, maximizes the number of matches, and then minimizes the sum of squared distances.

These are repeated PanNuke fold 1 development comparisons. Training images are used to fit the model; audit images compare training approaches. The two random seeds repeat each comparison under different training randomness; they do not change the data split. The audit contains 522 images, of which 11 contain 145 interior Dead nuclei. It is not a new independent patient cohort. Point matching is not instance segmentation PQ or an official three-fold test result. Each new approach is compared with its own freshly trained, same-seed control, rather than ranked across different experimental rounds.

2 What P17–P24 established

P17–P24 asked whether different image features, training settings, or model updates could improve localization. None of these approaches met all prespecified decision criteria in both seeds; their results reveal different kinds of difficulty.

Training a convolutional model from scratch (P17). Fit-set overall recall was only 14.41% and 11.92% in the two seeds, and audit coverage was also low. Architecture, pretraining, initialization and optimization changed together, so the comparison evaluates the combined approach.

Adding RGB information and changing batch size (P18–P19). P18 retained UNI and added an RGB residual path; Dead recall and precision remained below their minimum requirements. P19 increased effective batch size from 16 to 32: Dead hits changed from 66 to 57 and from 63 to 63. The larger batch also halved updates per epoch, so this compares training behavior as well as batch size.

Enlarging the input and adding an auxiliary task (P20–P21). P20 interpolated 256-pixel images to 512 before encoding. Dead hits rose from 66 to 79 and 69 to 75, while recall and precision remained below their minimum requirements and training cost rose considerably. Interpolation does not create new image detail. P21 added five class-specific center maps as training-only auxiliary targets; its Dead hits changed from 64 to 65 and 70 to 65.

Depth, parameter averaging, and learning rate (P22–P24). P22 trained all 24 encoder blocks and gained a net 18 and 20 Dead hits. Object-level analysis showed both gains and losses. P23's exponential moving average gained one and three hits. P24 increased the uniform learning rate of the last 12 blocks from 10^{-5} to 3×10^{-5}. Precision rose in both seeds, but Dead hits changed 67 to 78 and 68 to 66. Small Dead hits improved from 7/32 to 11/32 in both seeds while changes among larger objects offset some of that gain.

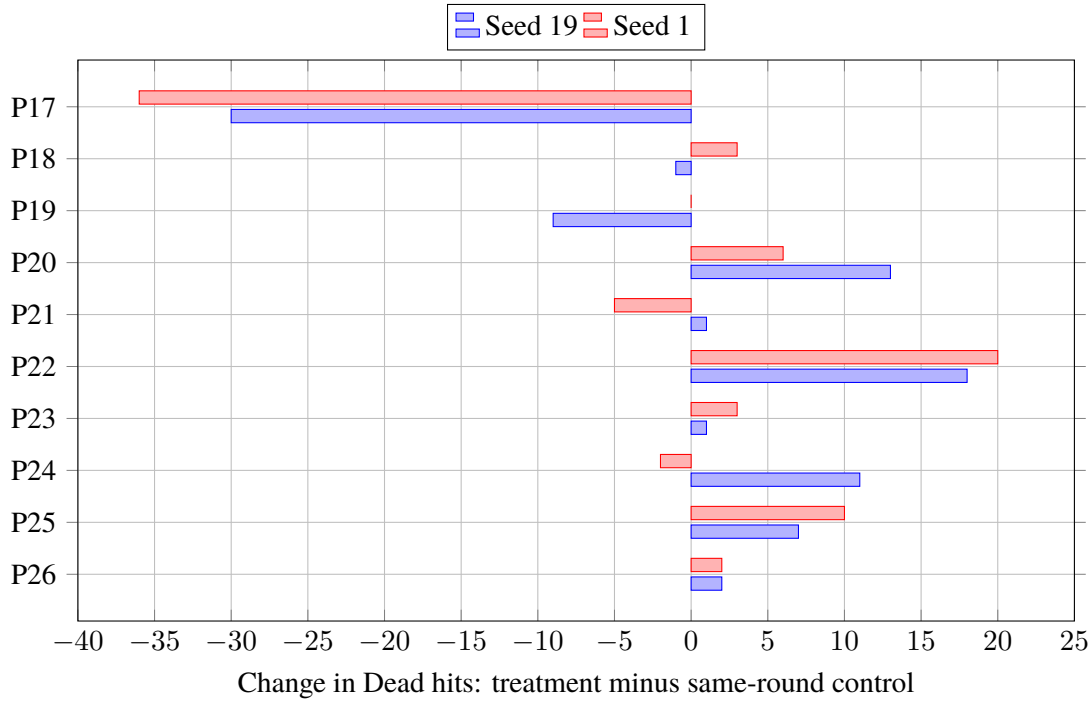


Figure 1: P17–P26: same-round paired Dead hit differences out of 145 audit objects. Each bar uses that round's own control, not a direct randomized comparison across rounds. P22 yields the largest consistent gains, but does not pass the full gate.

3 P25: full-depth layerwise adaptation

The encoder is the part of the model that extracts information from an image. Control C trains the last 12 UNI encoder blocks at 10^{-5}. New approach D trains all 24, with initial block-b rate $3 \times 10^{-5} \cdot 0.95^{23-b}$: approximately 9.22×10^{-6} at the bottom and 3×10^{-5} at the top. The data, 130-epoch schedule, loss, decoder and prespecified decision criteria remain fixed. D changes both trainable depth and the rate distribution; the comparison evaluates the combined configuration.

All four runs completed 130 epochs and 17,420 optimizer updates; results use the last checkpoint. Overall recall has 12,250 annotated nuclei as its denominator, and Dead recall has 145. Precision uses each run's actual candidate count.

Seed/arm	Dead hits	Dead recall	Overall recall	Precision	Small Dead/32
19 C	69	.475862	.589878	.690031	8
19 D	76	.524138	.650367	.744371	10
1 C	65	.448276	.598041	.691850	7
1 D	75	.517241	.639837	.742446	12

Table 1: P25 fixed-decoder audit results.

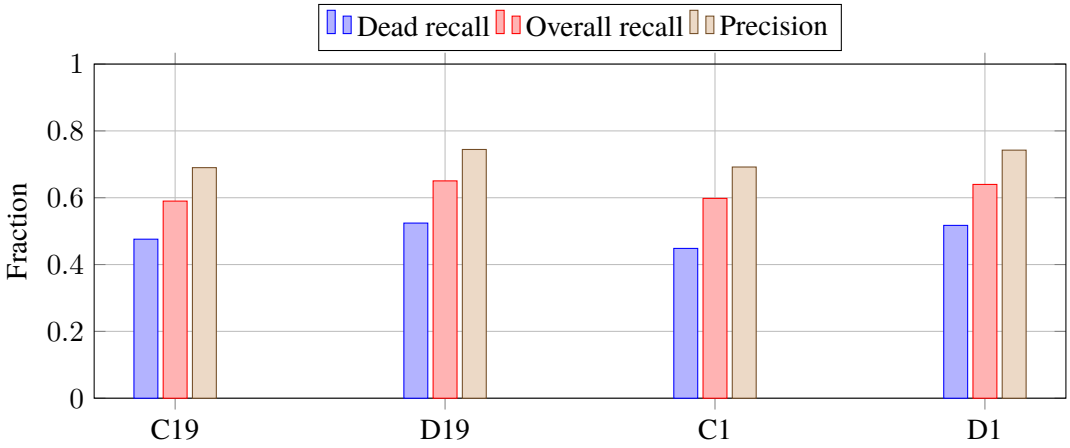


Figure 2: D improves both overall recall and precision in each seed, while Dead recall remains below the original 0.60 floor.

In seed 19, D matched 741 more nuclei (7,226 to 7,967) while producing 231 more candidates (10,472 to 10,703). In seed 1, it matched 512 more (7,326 to 7,838) with 32 fewer candidates (10,589 to 10,557). Both precision and overall recall improved together.

3.1 The original decision rule

Both seeds had to meet every requirement: Dead recall at least 0.60, a gain of at least 0.05 over the same-seed control, a strictly positive lower bound of the original paired bootstrap 95% interval, precision at least 0.75, and the support, relative-performance and nondegeneracy conditions. A pass on one condition cannot replace a failure on another.

The paired bootstrap resamples Dead-positive audit images within each tissue 1,000 times (random seed 20261018) to describe uncertainty on this image set; it is not independent patient-level validation. Seed 19 gained $7/145 = 0.048276$, with interval $[-0.010417, 0.164835]$; seed 1 gained $10/145 = 0.068966$, with interval $[0, 0.194031]$. Both runs fall short of the minimum Dead recall and precision; neither interval has a strictly positive lower bound. Seed 19 also falls short of the gain minimum. The zero lower bound for seed 1 is not positive. The original rule and independent recomputation agree: both seeds fail, so we will not pursue this exact combined configuration further.

3.2 Where the net changes occur

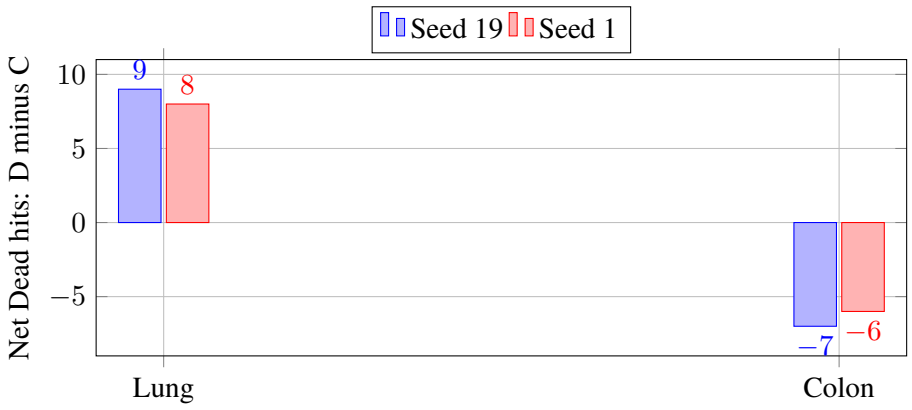


Figure 3: Net Dead hits by tissue: 60 in lung and 56 in colon. These comparisons were examined after the overall results.

The new approach detected nine and eight more Dead nuclei in lung images but seven and six fewer in colon images, respectively. In seed 19, the colon loss nearly offsets the lung gain. These are differences between per-image hit counts: a net gain of seven does not mean exactly seven previously missed objects were recovered while all others were retained. Five lung and colon images contain 116 of the 145 Dead nuclei, so a small number of images can have substantial influence.

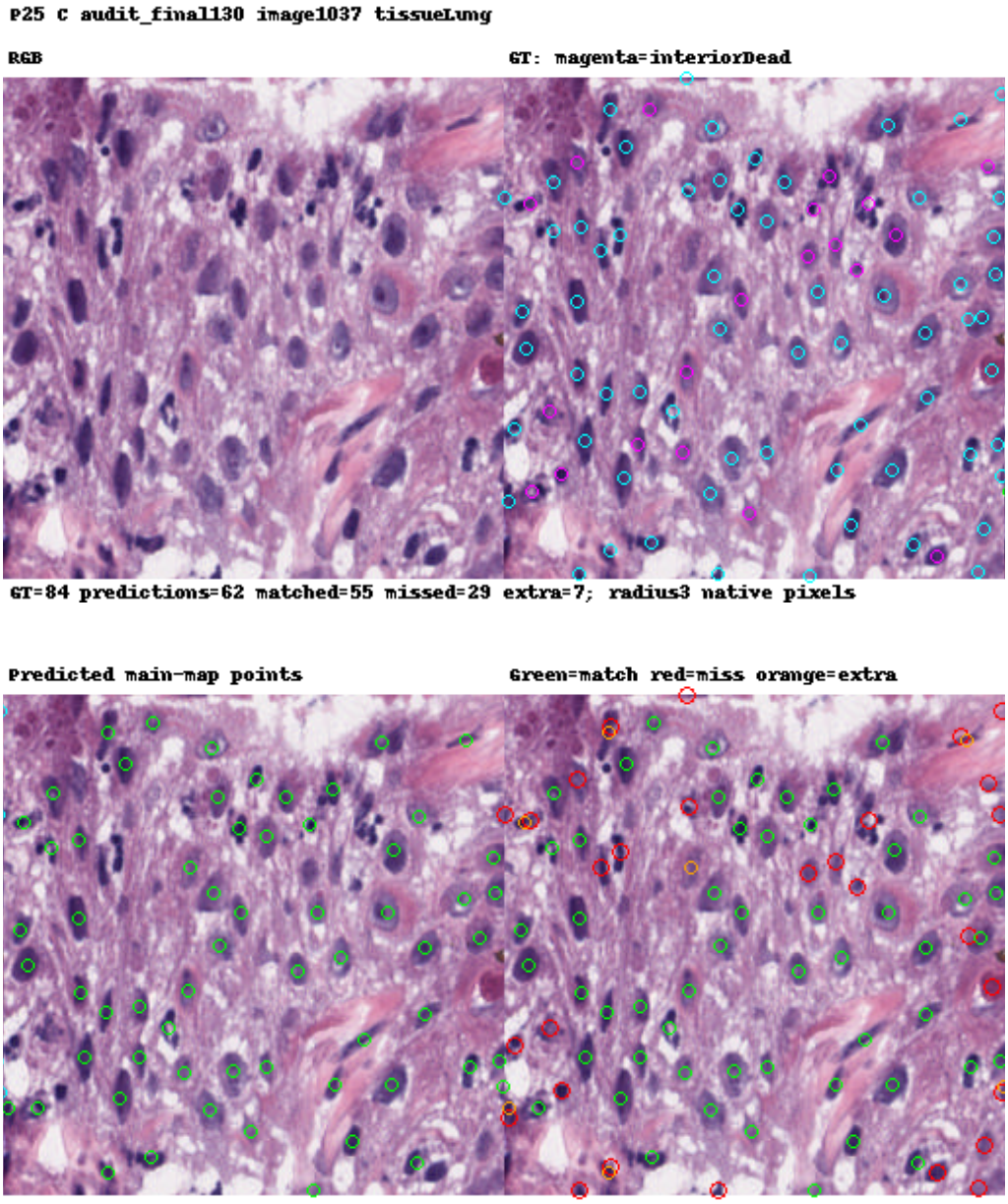


Figure 4: Seed 19 control C: image, annotations, predictions, and matching.

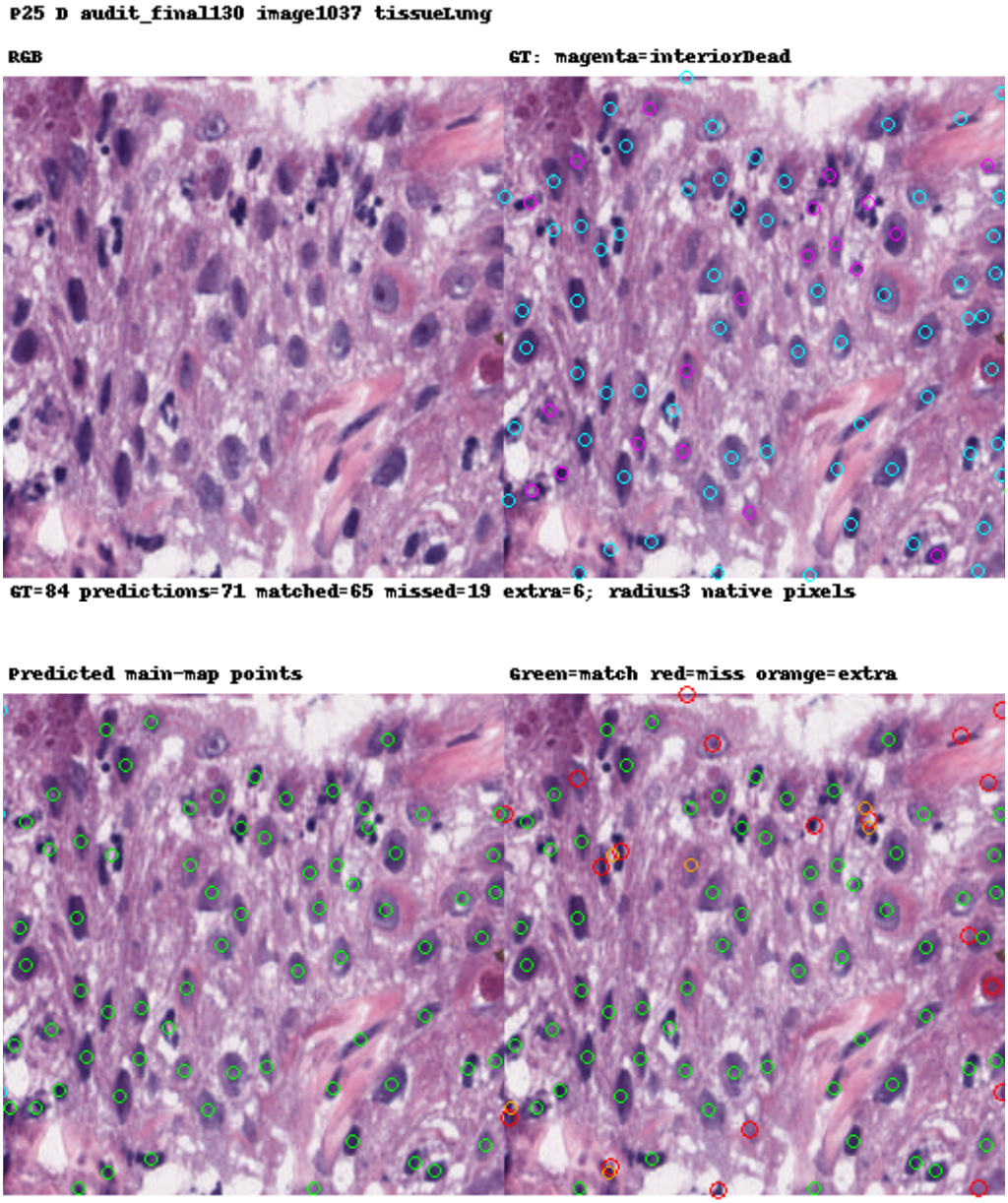


Figure 5: Predetermined audit image 1037, seed 19: D is shown here. The two rows show original image, annotated centers, predictions, and matching status. Magenta marks interior Dead annotations and cyan other annotations; green indicates a match, red a missed annotation, and orange an extra prediction. Image-level totals include all nuclei; the increase from 10 to 15 refers specifically to Dead nuclei. This one image illustrates, but cannot represent, the full cohort.

3.3 Fitting versus transfer

D finds 649/656 Dead nuclei on the fit images in both seeds, about 98.9%; on audit images it finds 76/145 and 75/145, about 52%. The model fits the training images closely, but performance is much lower on the audit images. Image differences, model features, the training objective and optimization are possible contributors.

4 P26: testing an anchor to pretrained weights

P26 uses a same-seed comparison of C and R. The added penalty encourages the trainable encoder parameters to remain near their pretrained values while allowing them to change. R adds only this L2SP term:

$$L_R = L_C + \frac{0.01}{2} \sum_{\theta \in \text{trainable encoder}} (\theta - \theta_0)^2.$$

The detection head is not anchored. The original AdamW weight decay of 0.01 remains; the penalty is applied once per effective batch. Both arms train the last 12 blocks for 130 epochs with seeds 19 and 1, the last checkpoint and the unchanged decision criteria. The fit-audit gap does not itself show that pretrained knowledge was forgotten. Li et al. introduced L2SP at ICML 2018: `https://proceedings.mlr.press/v80/li18a.html`.

The final P26 CPU regression passed 1,096 tests with three skipped. V100 validation job 55661530 completed with COMPLETED/0:0 in 689 seconds, including 97 passing node checks. Validation compared both control seeds' gradients, parameters, optimizer and random states, and checked the added penalty's gradients and updates, BF16, frozen states and checkpoint reloads. A 1.5-fold projection for 130 epochs was 7.07/7.20 hours for C/R; peak training memory was 4.09/4.66 GiB. Hash verification covered 490 sealed source files and 29 small validation artifacts.

All four P26 tasks completed 130 epochs with SLURM COMPLETED/0:0. For seed 19, C and R matched 67 and 69 of 145 Dead nuclei; for seed 1, 64 and 66. In each seed the Dead gain was 0.013793. Paired 95% intervals were $[-0.006173, 0.053793]$ and $[-0.071429, 0.061224]$. R overall recall was 0.570857/0.569224 versus C 0.583673/0.596245, and R precision was 0.678537/0.688080 versus C 0.697221/0.692519. Both seeds fail the original gate, independently reproduced. Summary, completion and evaluation payloads for all four runs match their preservation manifests; checkpoints were not independently rehashed.

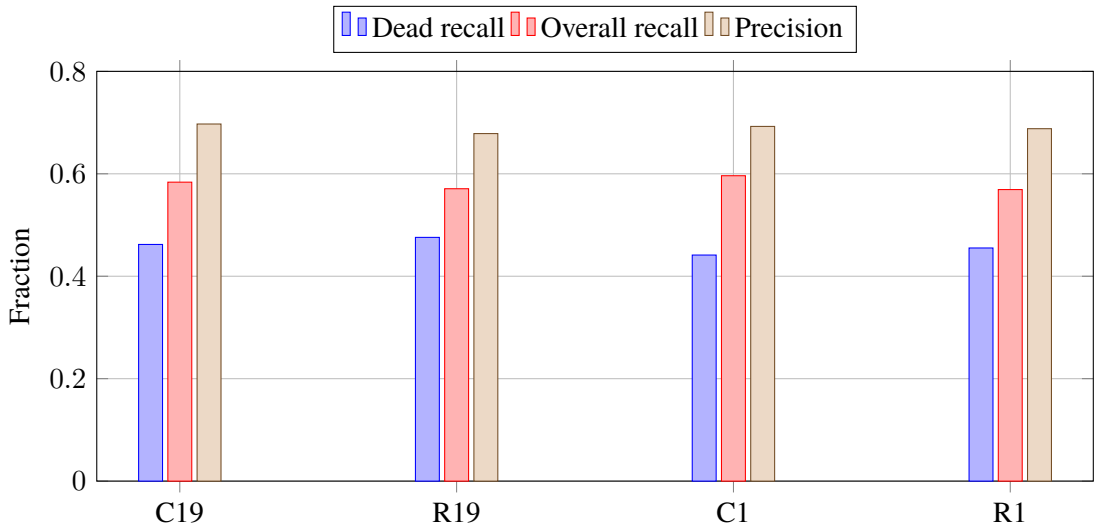


Figure 6: P26 four-run audit: the anchored arm gains two Dead matches in each seed while overall recall and precision fall.

A The paired comparisons across rounds

Each cell lists same-round control C followed by treatment, as Dead nuclei detected / precision. The Dead denominator is 145. P23 E is an exponential moving average of the same training trajectory, not another independent run. None of the treatments passed all original criteria in both seeds; the last column identifies the principal reason.

Round	Change	Seed 19 C → treatment	Seed 1 C → treatment	Principal failure
P17	CNN trained from scratch	60/.684 → 30/.952	67/.695 → 31/.949	Severe coverage loss
P18	RGB residual branch	71/.691 → 70/.694	64/.696 → 67/.696	Dead, precision, gain and interval
P19	Effective batch 16 to 32	66/.695 → 57/.689	63/.694 → 63/.695	Dead, precision, gain and interval
P20	Interpolate 256 to 512	66/.691 → 79/.712	69/.696 → 75/.724	Dead recall and precision minima
P21	Class-center auxiliary task	64/.692 → 65/.699	70/.696 → 65/.697	Dead, precision, gain and interval
P22	All 24 blocks, uniform rate	67/.692 → 85/.728	62/.699 → 82/.734	Dead recall and precision minima
P23	Weight averaging	69/.692 → 70/.694	60/.694 → 63/.691	Dead, gain and precision
P24	Higher rate, last 12 blocks	67/.688 → 78/.732	68/.694 → 66/.731	Dead recall and precision minima
P25	All blocks, layerwise rates	69/.690 → 76/.744	65/.692 → 75/.742	Dead recall, precision and interval
P26	Pretrained-weight anchor	67/.697 → 69/.679	64/.693 → 66/.688	Dead, precision, gain and interval

Table 2: Each round uses its own fresh control; precision rounded to three decimals.

P22 treatment finds 85 and 82 Dead nuclei, versus 76 and 75 for P25. These are not randomized treatments within a single paired experiment: P22 controls find 67 and 62, whereas P25 controls find 69 and 65. Training trajectories and layerwise rates differ. P22 uses a uniform rate across all blocks; P25 uses layerwise rates. The lower P25 counts therefore do not isolate which change caused the difference or establish a confirmed ranking. P25's improvement across all nuclei also does not automatically make it a better rare-Dead strategy: overall recall has 12,250 nuclei in its denominator, but the target subgroup has only 145.

B P25 four-run detail

The fit set contains 50,968 nuclei, including 656 interior Dead and 168 small Dead; the audit contains 12,250, 145 and 32, respectively. Extra points equal candidates minus matches. Training seconds sum the logged training portions; peak memory is measured during training, not the whole job.

Measure	C19	D19	C1	D1
Fit overall recall	.976162	.991701	.980576	.993388
Fit Dead /656	627	649	629	649
Fit small Dead /168	148	163	148	162
Audit overall recall	.589878	.650367	.598041	.639837
Audit Dead /145	69	76	65	75
Audit small Dead /32	8	10	7	12
Audit matched /12250	7226	7967	7326	7838
Audit candidates	10472	10703	10589	10557
Audit extra points	3246	2736	3263	2719
Training seconds	14384	20306	14365	20371
Peak memory GiB	4.047	6.745	4.047	6.745

Table 3: Last-checkpoint fit and audit summaries and logged training cost.

D produces 510 and 544 fewer extra points while matching 741 and 512 more nuclei. Its logged training time is approximately 1.41 and 1.42 times C; peak memory rises by about 2.70 GiB. Job-level elapsed time

also includes initialization, evaluation and saving. Shared storage and different nodes affect elapsed time, so the full difference cannot be assigned to model computation.

C All 11 positive images and fixed-example transitions

Image	Tissue	Dead	Seed 19 C→D	Seed 1 C→D
178	Breast	1	0→0	0→1
776	Colon	7	4→3	5→3
1025	Lung	25	18→19	19→18
1037	Lung	20	10→15	10→16
1038	Lung	15	9→12	9→12
1482	Colon	49	15→9	14→10
1756	Cervix	15	6→9	5→8
2076	Esophagus	8	6→7	1→6
2482	Stomach	3	0→0	0→0
2485	Stomach	1	0→1	1→0
2587	Uterus	1	1→1	1→1

Table 4: All Dead-positive audit images; entries are per-image hit counts.

Omitting each positive image in turn gives D-minus-C Dead recall changes from $+.016$ to $+.135417$ for seed 19 and $+.032$ to $+.145833$ for seed 1. This is a post-result influence description, not a new confidence interval or replacement for the original gate. The full 11-image differences remain $+.048276$ and $+.068966$.

The original bootstrap samples images containing interior Dead nuclei within each tissue. Four of the seven tissues (Breast, Cervix, Esophagus and Uterus) have only one positive image each, so those single-image groups do not vary under within-tissue resampling. The intervals describe variation available among the existing tissues and images.

The eight predetermined audit examples contain 53/145 Dead nuclei. Matching the same annotated objects across arms gives, for seed 19, 26 retained, 11 gained, six lost and ten missed by both. For seed 1 the counts are 30, eight, four and 11. Each row sums to 53 and per-image hit counts agree with the original saved scores. The sample net gains of five and four differ from the full audit gains of seven and ten. These examples explain specific object transitions; they do not estimate their frequency over the entire cohort.

D Evidence and verification

All four P25 top-level SLURM tasks report COMPLETED/0:0. Their elapsed times are 4:07:51, 5:49:40, 4:08:06 and 5:53:40. With one GPU per task, their sum is 19:59:17, or 19.9881 GPU hours, including job-level overhead. The collection records verification of 114 small files, including 48 final PNGs, and 485 sealed source files. Large checkpoints were checked against recorded sizes and preservation inventories; they were not independently transferred and rehashed in this review. The accompanying `NUMBER_SOURCES.md` maps the figures and claims to their records.